MiAnemone, a set of universal PCR primers for metabarcoding environmental DNA from sea anemones

Fei Xia

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Abstract

We designed and evaluated 22 primer pairs in silico for mitochondrial 12S and 16S rRNA genes in Actiniaria. In a 12S reference library of 250 sea anemone species, MD covered 224 of 242 evaluable species and matched both binding sites exactly in 208. In a 16S library of 265 species, LD covered 231 of 258 evaluable species, with 180 exact matches. Under strict Watson–Crick pairing, MD and LD matched nine and zero records in their respective background panels.

Keywords · Actiniaria, environmental DNA, mitochondrial rRNA, primer degeneracy, metabarcoding

中文摘要

本研究针对海葵目（Actiniaria）的线粒体 12S 和 16S rRNA 基因设计并计算评估了 22 对引物。在 250 种海葵的 12S 参考库中，MD 覆盖 224/242 个可评价物种，208 个物种的双端均完全匹配。在 265 种海葵的 16S 参考库中，LD 覆盖 231/258 个可评价物种，其中 180 个双端完全匹配。严格 Watson–Crick 模型下，两者在各自背景库中分别命中 9 条和 0 条记录。

1. Introduction

Primer design for environmental DNA metabarcoding must account for amplicon length, variation at binding sites among target species and matches to related non-target taxa. We drew on the conserved-flank design of MiFish and the use of degenerate sites in Anth-28S (Miya et al., 2015, McCartin et al., 2024). We compared ordinary and degenerate primers at identical coordinates in five size windows (XS–XL) per mitochondrial marker to determine how gains in coverage affect exact matching and background hits.



2. Materials and methods

2.1 Reference panels and primer design

Each target panel contains one GenBank sequence per named species. All coordinates are 1-based, inclusive positions in the respective ungapped mitochondrial rRNA gene of Metridium senile (NC_000933.1). Ordinary primers were selected from observed templates and consensus motifs following the conserved-flank approach of MiFish. D candidates incorporate lineage variation following the Anth-28S approach, with at most eight variants per oligo and ambiguity confined to its 3′ half.

2.2 Target coverage and non-target matching

Evaluable references span both sites and have no ambiguous nucleotides in their aligned binding regions. Both strands were searched for 100–500 bp products with at least 50 internal bases. Coverage allows two mismatches per oligo with compatibility at each of its final three bases. Exact matches require compatibility at every base on both ends.

Background scans allow 0–1,000 internal bases. Strict WC and G–T models allow two mismatches per oligo and require compatibility across its last three bases. Relaxed WC and G–T models allow three and two mismatches respectively without a terminal constraint. The selection thresholds were at most ten hits per strict model and 50 per relaxed model.

2.3 Base-wise compatibility

For each primer end, we retained evaluable species with an unambiguous, contiguous aligned binding site of the reference length. Insertions and deletions within that end were excluded. We reverse-complemented reverse sites so all primer bases are read in synthesis orientation. Each site percentage is the fraction of retained species whose template base belongs to the primer's IUPAC base set. This site-specific denominator can differ between ends. Paired coverage was evaluated separately by searching both strands of the raw sequences.

2.4 Reference insert identity

We removed both primer sites from the homologous interval and counted unambiguous inserts unique to one species within each window panel. A unique reference insert measures discrimination by sequence identity in the sampled panel.

2.5 Independent verification

Independent paired-hit checks on target and background sequences agreed across 8,940 record–model comparisons for 12S and 45,780 for 16S. Model counts use records in the corresponding marker panel.

3.1 Results · mitochondrial 12S

The 12S panel comprises 250 target species and 322 non-target cnidarian species. MD covered 224/242 species with 208 exact matches. Both M and MD hit nine background records under strict WC pairing.

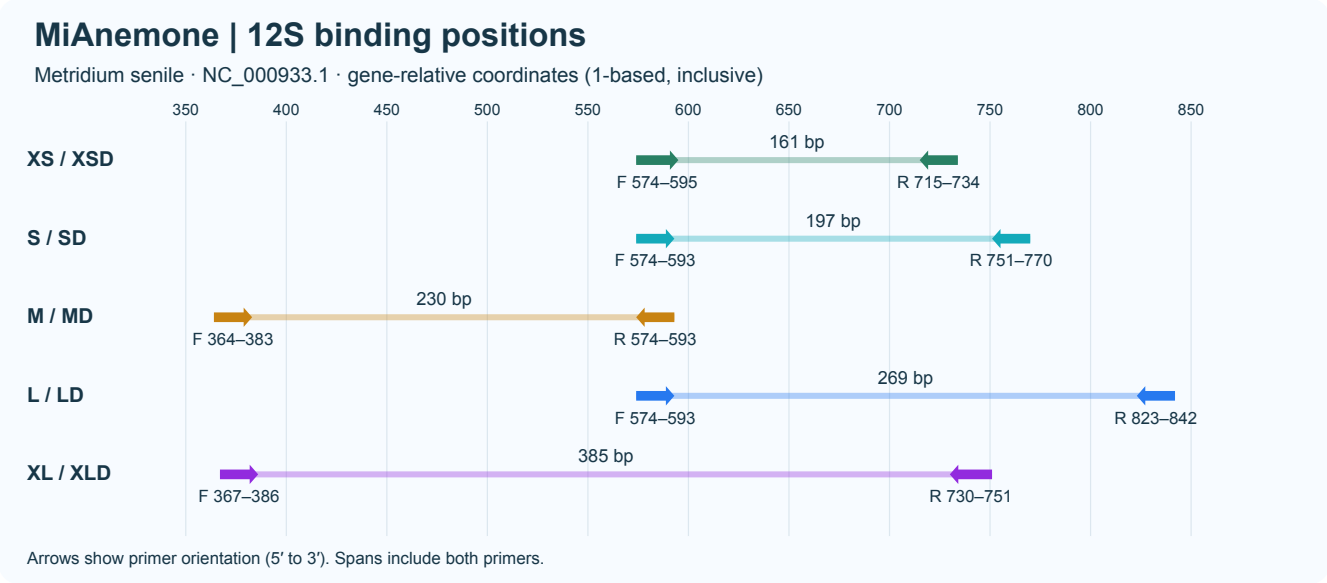


Figure 1. 12S reference binding sites on NC_000933.1. Arrows point in the synthesis direction. Spans include both primers.

Table 1. Predicted paired matches among evaluable target species (N). Background cells show WC / G–T record counts for each mismatch rule.

Pair	N	Covered	Exact	Strict background	Relaxed background
XS	245	205	2	9 / 9	9 / 9
XSD	245	230	0	9 / 9	9 / 9
S	246	173	0	0 / 0	0 / 0
SD	246	195	26	0 / 0	0 / 0
M	242	220	198	9 / 9	9 / 9
MD	242	224	208	9 / 9	9 / 9
L	246	226	1	9 / 9	9 / 9
LD	246	229	1	9 / 9	9 / 9
XL	243	185	46	0 / 0	1 / 0
XLD	243	225	52	0 / 0	18 / 0

3.2 Results · mitochondrial 16S

The 16S panel comprises 265 target species and 1,775 non-target records. LD covered 231/258 species with 180 exact matches. H and HD covered 146/165 and 154/165 at the original site, but both exceeded the relaxed background thresholds.

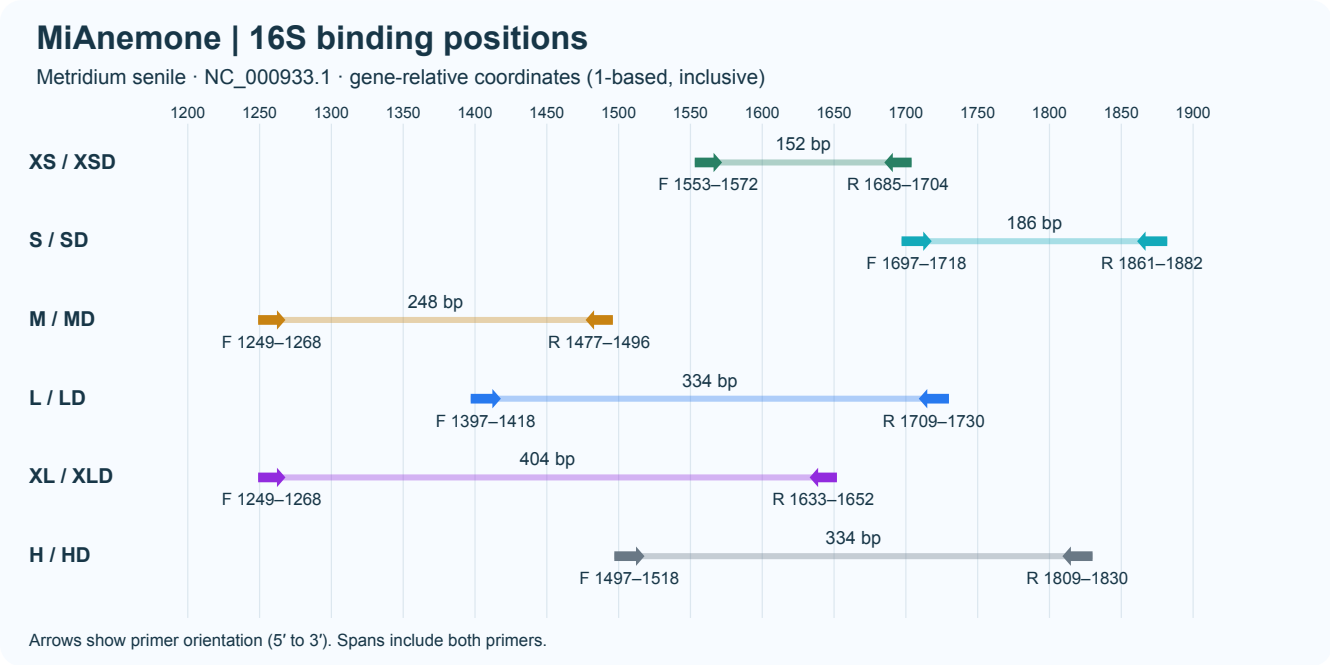


Figure 2. 16S reference binding sites on NC_000933.1. Arrows point in the synthesis direction. Spans include both primers.

Table 2. Predicted paired matches among evaluable target species (N). Background cells show WC / G–T record counts for each mismatch rule.

Pair	N	Covered	Exact	Strict background	Relaxed background
H	165	146	122	0 / 0	486 / 500
HD	165	154	136	0 / 1	976 / 501
XS	262	165	132	0 / 0	0 / 0
XSD	262	223	1	0 / 0	1 / 0
S	77	63	0	0 / 0	0 / 0
SD	77	70	56	0 / 0	0 / 0
M	76	61	51	0 / 0	0 / 0
MD	76	61	52	0 / 0	0 / 0
L	258	138	106	0 / 0	7 / 4
LD	258	231	180	0 / 3	46 / 10
XL	76	41	36	0 / 0	0 / 0
XLD	76	61	52	0 / 0	0 / 0

3.3 Results · 12S base matching

12S | Primer–template compatibility by nucleotide

5′ → 3′ in each row. Cells show the primer base and percent compatible; the final three bases are outlined.

XS / XSD

F 574–595 R 715–734

XS F	n=230	C 98	G 99	A 100	A 100	A 100	G 100	G 100	T 98	A 97	G 1	A 100	A 100	T 100	C 100	T 100	T 100	C 100	C 100	G 100	C 100	C 100	T 100
XSD F	n=230	T 2	G 99	A 100	A 100	A 100	G 100	G 100	T 98	A 97	A 99	A 100	A 100	T 100	C 100	T 100	T 100	C 100	C 100	G 100	C 100	C 100	T 100
XS R	n=244	C 98	G 100	A 100	C 100	G 100	G 100	C 100	C 100	A 100	T 100	G 100	C 100	A 100	A 98	T 100	A 100	C 100	C 100	G 88	G 100		
XSD R	n=244	A 2	G 100	A 100	C 100	G 100	G 100	C 100	C 100	A 100	T 100	G 100	C 100	A 100	R 100	T 100	A 100	C 100	C 100	D 100	G 100		

S / SD

F 574–593 R 751–770

S F	n=231	C 98	G 99	A 100	A 100	A 100	G 100	G 100	T 98	C 1	A 99	A 100	A 100	T 100	C 100	T 100	T 100	C 100	C 100	G 100	C 100	
SD F	n=231	C 98	G 99	A 100	A 100	A 100	G 100	G 100	T 98	A 97	A 99	A 100	A 100	T 100	C 100	T 100	T 100	C 100	C 100	G 100	C 100	
S R	n=234	C 87	G 100	G 100	G 79	T 99	T 100	G 99	G 80	G 99	T 98	C 97	C 100	C 100	T 98	T 100	T 21	T 100	A 97	G 88	C 78	
SD R	n=234	C 87	G 100	G 100	A 20	T 99	T 100	G 99	G 80	G 99	T 98	C 97	C 100	C 100	T 98	T 100	Y 100	T 100	A 97	G 88	Y 100	

M / MD

F 364–383 R 574–593

M F	n=235	A 100	A 97	G 100	G 100	C 99	G 98	T 99	A 99	A 97	A 100	G 100	G 100	A 100	T 100	G 100	T 100	G 100	T 100	A 97	G 100	
MD F	n=235	A 100	A 97	G 100	G 100	C 99	G 98	T 99	A 99	A 97	A 100	G 100	G 100	A 100	T 100	G 100	T 100	G 100	T 100	V 100	G 100	
M R	n=227	G 100	C 100	G 100	G 100	A 100	A 100	G 100	A 100	T 100	T 100	T 99	T 96	A 98	C 100	C 100	T 100	T 100	T 100	C 99	G 98	
MD R	n=227	G 100	C 100	G 100	G 100	A 100	A 100	G 100	A 100	T 100	T 100	T 99	D 100	A 98	C 100	C 100	T 100	T 100	T 100	C 99	R 100	

L / LD

F 574–593 R 823–842

L F	n=231	T 2	G 99	A 100	A 100	A 100	G 100	G 100	T 98	A 97	A 99	A 100	A 100	T 100	C 100	T 100	T 100	C 100	C 100	G 100	C 100	
LD F	n=231	T 2	G 99	A 100	A 100	A 100	G 100	G 100	T 98	A 97	A 99	A 100	A 100	T 100	C 100	T 100	T 100	C 100	C 100	G 100	C 100	
L R	n=245	T 100	A 100	T 100	A 99	A 99	G 93	A 99	A 98	A 100	A 99	T 100	G 100	T 100	A 100	G 100	C 100	G 100	C 100	A 100	T 99	
LD R	n=245	T 100	A 100	T 100	A 99	A 99	G 93	A 99	A 98	A 100	A 99	T 100	G 100	T 100	A 100	G 100	C 100	G 100	C 100	A 100	Y 100	

XL / XLD

F 367–386 R 730–751

XL F	n=235	G 100	C 99	G 99	T 100	A 99	A 97	A 100	G 100	G 100	A 100	T 100	G 100	T 100	G 100	T 100	A 97	G 98	G 100	C 94	C 86	
XLD F	n=235	G 100	C 99	G 99	T 100	A 99	A 97	A 100	G 100	G 100	A 100	T 100	G 100	T 100	G 100	T 100	A 97	R 100	G 100	Y 100	Y 100	
XL R	n=237	C 78	T 100	T 98	A 96	A 100	C 47	A 98	C 100	A 98	C 95	G 99	G 48	A 97	C 99	T 99	T 100	A 100	C 98	G 100	A 100	C 100	G 100
XLD R	n=237	C 78	T 100	T 98	A 96	A 100	C 47	A 98	C 100	A 98	C 95	G 99	R 100	A 97	C 99	T 99	T 100	A 100	C 98	G 100	A 100	C 100	G 100

Figure 3. Each cell gives the synthesis base and rounded percentage of compatible reference species. Rows are ordinary and D forward ends followed by ordinary and D reverse ends. Colors mark <50%, 50–<90%, 90–<100% and 100% compatibility. The 3′ triplet is outlined. Counts for the two ends may differ. Exact numerators and denominators are provided with the figure data.

3.4 Results · 16S base matching

16S | Primer–template compatibility by nucleotide

5′ → 3′ in each row. Cells show the primer base and percent compatible; the final three bases are outlined.

XS / XSD

F 1553–1572 R 1685–1704

XS F	n=248	A 100	G 100	T 98	T 100	T 100	A 100	G 100	A 100	T 99	A 97	A 100	T 97	G 100	T 100	G 100	G 100	G 93	A 93	T 71	C 100
XSD F	n=248	A 100	G 100	T 98	T 100	T 100	A 100	G 100	A 100	T 99	A 97	A 100	T 97	G 100	T 100	G 100	G 100	A 5	R 100	B 98	C 100
XS R	n=249	T 100	A 99	A 99	C 87	T 97	A 100	C 98	G 96	G 95	A 98	C 100	C 100	T 98	T 100	A 100	A 100	G 100	C 98	T 97	C 100
XSD R	n=249	T 100	A 99	A 99	T 12	T 97	A 100	C 98	G 96	G 95	A 98	C 100	C 100	T 98	T 100	A 100	A 100	G 100	C 98	T 97	C 100

S / SD

F 1697–1718 R 1861–1882

S F	n=74	C 96	G 96	T 100	A 96	G 91	T 99	T 100	A 100	A 100	T 100	A 100	G 93	C 100	A 15	A 31	G 96	T 96	G 100	G 99	C 100	C 97	T 100
SD F	n=74	C 96	G 96	T 100	A 96	G 91	T 99	T 100	A 100	A 100	T 100	A 100	G 93	C 100	H 100	R 93	G 96	T 96	G 100	G 99	C 100	C 97	T 100
S R	n=76	A 100	G 100	G 100	T 100	C 100	G 100	C 100	A 100	A 100	A 100	C 100	A 96	T 96	C 97	G 100	T 95	T 5	G 97	T 100	C 97	G 92	A 100
SD R	n=76	A 100	G 100	G 100	T 100	C 100	G 100	C 100	A 100	A 100	A 100	C 100	A 96	T 96	C 97	G 100	K 100	Y 100	G 97	T 100	C 97	R 100	A 100

M / MD

F 1249–1268 R 1477–1496

M F	n=70	A 99	A 97	G 90	C 90	T 96	A 91	A 100	A 96	T 81	G 99	C 89	T 91	G 100	A 96	A 94	G 99	T 100	A 99	T 100	G 96
MD F	n=70	A 99	A 97	G 90	C 90	T 96	A 91	A 100	A 96	T 81	G 99	C 89	T 91	G 100	A 96	A 94	G 99	T 100	A 99	T 100	G 96
M R	n=74	T 100	T 100	C 100	G 100	T 100	C 100	T 100	T 96	A 100	C 100	A 100	A 97	T 100	A 89	T 100	G 100	G 96	A 100	C 100	G 100
MD R	n=74	T 100	T 100	C 100	G 100	T 100	C 100	T 100	T 96	A 100	C 100	A 100	A 97	T 100	W 100	T 100	G 100	D 100	A 100	C 100	G 100

L / LD

F 1397–1418 R 1709–1730

L F	n=254	G 100	G 99	C 100	C 98	G 100	G 100	T 100	A 100	T 100	G 100	A 100	A 100	T 98	G 100	G 100	C 100	G 69	T 98	C 89	A 100	C 100	G 100
LD F	n=254	G 100	G 99	C 100	C 98	G 100	G 100	T 100	A 100	T 100	G 100	A 100	A 100	T 98	G 100	G 100	C 100	R 100	Y 100	Y 100	A 100	C 100	G 100
L R	n=248	C 96	C 100	C 100	C 100	T 100	T 92	G 88	C 100	A 100	G 100	T 100	C 100	A 99	G 98	G 99	C 99	C 99	A 96	C 96	C 70	G 56	G 98
LD R	n=248	C 96	C 100	C 100	C 100	T 100	T 92	G 88	C 100	A 100	G 100	T 100	C 100	A 99	G 98	G 99	C 99	C 99	A 96	C 96	Y 96	D 99	G 98

XL / XLD

F 1249–1268 R 1633–1652

XL F	n=70	A 99	A 97	G 90	C 90	T 96	A 91	A 100	A 96	T 81	G 99	C 89	T 91	G 100	A 96	A 94	G 99	T 100	A 99	T 100	G 96
XLD F	n=70	A 99	A 97	G 90	C 90	T 96	A 91	A 100	A 96	T 81	G 99	C 89	T 91	G 100	A 96	A 94	G 99	T 100	A 99	T 100	G 96
XL R	n=76	A 100	A 97	C 99	C 99	A 99	A 99	A 99	C 99	T 99	A 93	T 97	C 99	C 87	A 99	A 100	C 100	T 99	T 88	A 55	C 99		
XLD R	n=76	A 100	A 97	C 99	C 99	A 99	A 99	A 99	C 99	T 99	A 93	T 97	C 99	Y 95	A 99	A 100	C 100	T 99	Y 99	R 97	C 99		

H / HD

F 1497–1518 R 1809–1830

H F	n=163	A 99	A 100	G 100	A 94	C 100	C 100	C 100	C 100	A 97	T 100	T 97	G 100	A 100	G 100	C 100	T 100	T 100	T 100	A 100	C 100	T 100	A 96
HD F	n=163	A 99	A 100	G 100	A 94	C 100	C 100	C 100	C 100	A 97	T 100	T 97	G 100	A 100	G 100	C 100	T 100	T 100	T 100	A 100	C 100	Y 100	W 100
H R	n=155	C 100	C 100	C 100	C 100	A 98	T 97	G 100	G 100	T 98	A 96	G 97	C 98	T 100	T 99	T 100	T 100	A 100	T 100	T 89	C 99	G 99	T 100
HD R	n=155	C 100	C 100	C 100	C 100	A 98	T 97	G 100	G 100	T 98	A 96	G 97	C 98	T 100	T 99	T 100	T 100	A 100	T 100	Y 100	C 99	R 100	T 100

Figure 4. Each cell gives the synthesis base and rounded percentage of compatible reference species. Rows are ordinary and D forward ends followed by ordinary and D reverse ends. Colors mark <50%, 50–<90%, 90–<100% and 100% compatibility. The 3′ triplet is outlined. Counts for the two ends may differ. Exact numerators and denominators are provided with the figure data.

3.5 Results · individual template alignments

The alignments below show the observed template bases in primer orientation. Each excerpt uses one window-evaluable accession from each of the first eight represented families, ordered by family, species and accession. These rows illustrate binding-site variation. Panel-wide counts are reported in Tables 1 and 2.

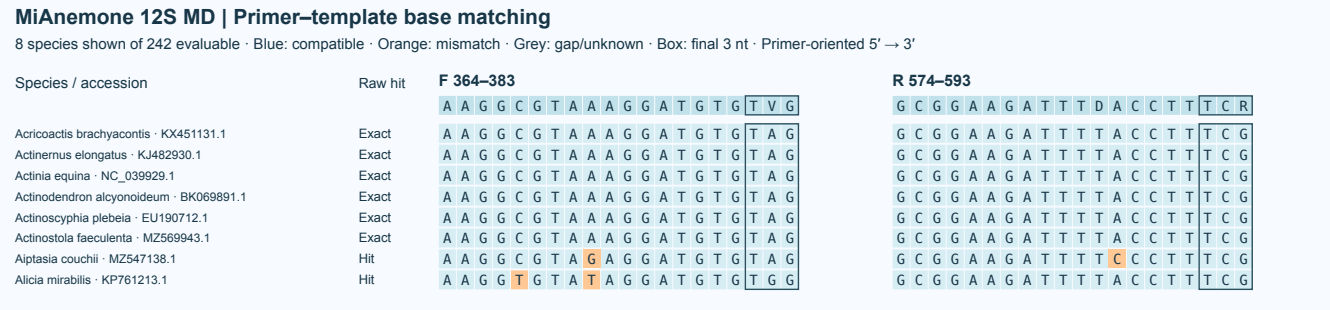


Figure 5. 12S MD. Blue cells indicate compatible bases, orange cells indicate mismatches and grey cells indicate gaps or unknown bases. Boxes mark the terminal triplets.

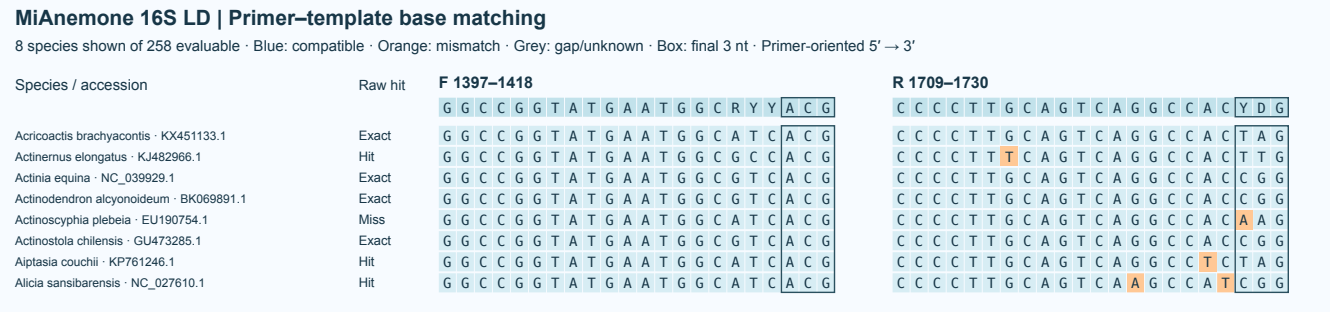


Figure 6. 16S LD. Both panels omit insertions relative to the reference. The raw-hit column reports the full-sequence paired search. Full-species alignments for all 22 pairs.

4. Discussion

12S MD yielded the most exact matches among the 12S pairs. Compared with ordinary M, it covered four additional species and provided ten additional exact matches with the same strict background count. Where a shorter amplicon and fewer background hits are priorities, the 197 bp SD pair covered 195/246 species with zero hits in all four background models. For 16S, LD covered 93 more species than ordinary L, while relaxed WC hits increased from seven to 46.

Shorter windows illustrate the distinction between coverage and exact matching. 12S XSD covered 230/245 species with zero exact matches. For 16S, XSD covered 58 more species than XS, but exact matches fell from 132 to one. Ordinary and D sequences were selected independently at each site, so both metrics matter when choosing a pair. Comparisons across windows also depend on how many references span both sites, particularly for the sparsely represented 16S windows. These results describe predicted template compatibility. Amplification efficiency in environmental samples remains to be measured.

The 12S XL window had unique inserts in 100/239 species with unambiguous inserts, compared with 35/244 for XS. In 16S, the H window had 87 unique inserts among 159 species. Different taxa contribute to each denominator, so these counts cannot isolate an effect of length alone.

Table 3. Primer-free reference insert identity by window.

Marker	Window	Unambiguous inserts	Unique to one species
12S	XS	244	35
12S	S	244	44
12S	M	239	86
12S	L	243	68
12S	XL	239	100
16S	H	159	87
16S	XS	259	58
16S	S	77	29
16S	M	75	31
16S	L	255	86
16S	XL	75	48

5. Primer sequences

Both ends are written in the 5′ to 3′ synthesis direction. Each D pair shares the four reference endpoints of its ordinary counterpart. Degeneracy counts the number of oligo variants per end.

Table 4. All 22 primer pairs in the released FASTA.

Marker	Pair	Forward (5′–3′)	Reverse (5′–3′)	Deg F/R
12S	XS	CGAAAGGTAGAATCTTCCGCCT	CGACGGCCATGCAATACCGG	1/1
12S	XSD	TGAAAGGTAAAATCTTCCGCCT	AGACGGCCATGCARTACCDG	1/6
12S	S	CGAAAGGTCAAATCTTCCGC	CGGGTTGGGTCCCTTTTAGC	1/1
12S	SD	CGAAAGGTAAAATCTTCCGC	CGGATTGGGTCCCTTYTAGY	1/4
12S	M	AAGGCGTAAAGGATGTGTAG	GCGGAAGATTTTACCTTTCG	1/1
12S	MD	AAGGCGTAAAGGATGTGTVG	GCGGAAGATTTDACCTTTCR	3/6
12S	L	TGAAAGGTAAAATCTTCCGC	TATAAGAAAATGTAGCGCAT	1/1
12S	LD	TGAAAGGTAAAATCTTCCGC	TATAAGAAAATGTAGCGCAY	1/2
12S	XL	GCGTAAAGGATGTGTAGGCC	CTTAACACACGGACTTACGACG	1/1
12S	XLD	GCGTAAAGGATGTGTARGYY	CTTAACACACGRACTTACGACG	8/2
16S	H	AAGACCCCATTGAGCTTTACTA	CCCCATGGTAGCTTTTATTCGT	1/1
16S	HD	AAGACCCCATTGAGCTTTACYW	CCCCATGGTAGCTTTTATYCRT	4/4
16S	XS	AGTTTAGATAATGTGGGATC	TAACTACGGACCTTAAGCTC	1/1
16S	XSD	AGTTTAGATAATGTGGARBC	TAATTACGGACCTTAAGCTC	6/1
16S	S	CGTAGTTAATAGCAAGTGGCCT	AGGTCGCAAACATCGTTGTCGA	1/1
16S	SD	CGTAGTTAATAGCHRGTGGCCT	AGGTCGCAAACATCGKYGTCRA	6/8
16S	M	AAGCTAAATGCTGAAGTATG	TTCGTCTTACAATATGGACG	1/1
16S	MD	AAGCTAAATGCTGAAGTATG	TTCGTCTTACAATWTGDACG	1/6
16S	L	GGCCGGTATGAATGGCGTCACG	CCCCTTGCAGTCAGGCCACCGG	1/1
16S	LD	GGCCGGTATGAATGGCRYYACG	CCCCTTGCAGTCAGGCCACYDG	8/6
16S	XL	AAGCTAAATGCTGAAGTATG	AACCAAACTATCCAACTTAC	1/1
16S	XLD	AAGCTAAATGCTGAAGTATG	AACCAAACTATCYAACTYRC	1/8

Data availability

Accession panels, alignments, oligos, per-species predictions and the scripts for verification and figure regeneration are archived in the MiAnemone repository.

References

1. Miya M et al. (2015). MiFish, a set of universal PCR primers for metabarcoding environmental DNA from fishes: detection of more than 230 subtropical marine species. Royal Society Open Science, 2, 150088. DOI

2. McCartin L et al. (2024). Nuclear eDNA metabarcoding primers for anthozoan coral biodiversity assessment. PeerJ, 12, e18607. DOI